Chapter 1: Differentiation

Original practice material for this MVP; not an official exam paper.

Chain rule: if y = [u(x)]^n, then dy/dx = n[u(x)]^(n-1) times du/dx.

Example: d/dx [(3x+1)^2] = 6(3x+1).

Product rule: d/dx [u(x)v(x)] = u prime times v + u times v prime.

Power rule: d/dx [a x^n] = a n x^(n-1).